Active Wind Rejection Control for a Quadrotor UAV Against Unknown Winds

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This article presents an active wind rejection control scheme for a quadrotor unmanned aerial vehicle (UAV) against unknown winds. Based on the estimated wind effects acting on the aircraft, the proposed control scheme can maintain the performance of the quadrotor UAV in the presence of model uncertainties, unknown winds, and system noises. Firstly, a two-stage particle filter is designed to estimate the UAV states and wind information from the motion of the vehicle without additional wind sensors. Then, an active wind rejection control scheme is proposed to actively attenuate the wind disturbances based on the estimated wind information. In the controller design, the nonsingular terminal sliding-mode control (NTSMC) is chosen as the baseline controller. To tackle the issues of model uncertainties and wind estimation errors, the adaptive drag coefficients are adopted to generate the compensation control signals. Finally, simulation results are presented to demonstrate the effectiveness of the proposed active wind rejection control scheme for a quadrotor UAV against unknown winds.

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I. INTRODUCTION

Wind behaviors have considerable impacts on unmanned aerial vehicles (UAVs) in a lot of outdoor tasks. The precise and agile control schemes must be considered to ensure the performance of UAVs in the strong and unpredictable wind environments [1], [2]. For example, when being employed in the applications of forest fire detection, monitoring, and fighting, UAVs require to closely track designate trajectories to collect fire information within the hazardous fire region against unknown winds. The adverse forces and moments generated by the external wind disturbances may degrade the performance of UAVs and even result in serious accidents if appropriate reactions are not promptly activated to attenuate the wind effects. Especially, quadrotor UAVs are more sensitive to those external disturbances due to their small sizes and light weights. Therefore, it is necessary to investigate the effective wind rejection control for UAVs to ensure their flight performance in the presence of unknown winds.

In recent years, lots of efforts have been invested in developing the control strategies of UAVs against wind disturbances. The robust control strategies are usually adopted to attenuate wind disturbances. However, robust control methods always have a high degree of conservativeness for the external disturbances, which are defined by norm bounded variables [3]. In [4], two robust nonlinear sliding-mode controllers are designed in the presence of wind perturbations. A switching model predictive attitude control is adopted for the quadrotor UAV against wind gust disturbances based on a piecewise affine model [5]. The $\mathcal{L}_1$ backstepping trajectory tracking control is developed to allow for the fast adaption to rapidly varying model uncertainties and wind disturbances [6]. In [7], a robust nonlinear dynamic inversion attitude controller is proposed in the presence of model uncertainties and wind gust disturbances, aerodynamic interactions with nearby structures and ground effects, while a feedback position controller is used to handle the lateral and vertical transitions. In [8], the finite frequency H_∞ control is used for the attitude stabilization, while the PID-based H_∞ loop shaping control is used for the linear motion in the presence of wind disturbances. An acceleration feedback enhanced H_∞ control is adopted against wind disturbances in [9].

To improve the performance of the UAV system against wind disturbances, unknown wind effects acting on the UAV dynamics should be estimated and considered in the generation of control signals [10]. To achieve this purpose, disturbance estimation and control techniques are required to be investigated, such as disturbance-observer-based control, active disturbance rejection control, embedded model control, and composite hierarchical antidisturbance control. Chen et al. [11] proposed a detailed review of disturbance observers. In [12], a high-gain residual observer is implemented in an embedded robust attitude control of a quadrotor UAV for online estimation and compensation of wind gust disturbances. In [13], a multiple observers-based

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antidisturbance control is applied for a quadrotor UAV. A composite disturbance observer and an extended state observer are used in the position loop to mitigate the payload and wind disturbances, while an extended state observer is used in the attitude loop to reject modeling uncertainties and wind disturbances. In [14], an attitude control for a UAV quadrotor is developed in the framework of active disturbance rejection control and embedded model control to estimate and reject the disturbances. In view of the idea of composite hierarchical antidisturbance control, an adaptive composite disturbance rejection control is developed, containing active disturbance rejection control and disturbance observer, for attitude control of the agricultural quadrotor UAV [15].

In addition, the deep-learning-based control approaches have become increasingly attractive in recent years. To improve the performance of the UAV against wind disturbances, the deep-learning-based approach is usually adopted to learn the unknown aerodynamic representation of the aircraft. In [16], a deep-learning-based trajectory tracking controller is developed to learn for quickly adapting to rapidly changing wind conditions. To this end, the domain adversarially invariant meta-learning algorithm is developed for the offline learning of the common aerodynamic representation, while a composite adaption law is used to update wind-specific linear coefficients.

However, to the best of the authors' knowledge, the control accuracy of a quadrotor UAV should still be investigated in the presence of strong winds. The precise control of a quadrotor UAV in rapidly changing wind environments requires an accurate and prompt perception of the ambient winds [17], [18]. In most cases, the high-frequency and low-latency observers are used to estimate the lumped disturbance of the quadrotor UAV, including the model uncertainties and external wind disturbances, but they may not take quick and accurate responses to the rapidly changing winds. Therefore, the existing control strategies may not ensure the accuracy of the stationary and transient control of a quadrotor UAV against strong winds. In addition, the system noises of a quadrotor UAV cannot be zero in the practical engineering applications. The existing control strategies may not work perfectly or even diverge in consideration of the stochastic system noises.

The sliding-mode control method is often adopted to maintain the tracking performance of a quadrotor UAV, due to its inherent robustness to model uncertainties and disturbances. In [19], an adaptive fault-tolerant control based on a multivariable integral terminal sliding-mode control (TSMC) is developed in the presence of parametric uncertainties and actuator faults. An adaptive super-twisting sliding-mode control is adopted for a quadrotor UAV to track a desired trajectory under gust winds [20]. An adaptive backstepping global sliding-mode control is offered for a quadrotor UAV in the existence of model uncertainty, wind perturbation, and input saturation [21]. In [22], the position and attitude control of a quadrotor UAV is realized by an optimized fuzzy-based sliding-mode control.

Compared with the conventional SMC, the TSMC can offer finite-time convergence. However, the TSMC schemes have the singularity problem. To eliminate the singularity and guarantee the global finite-time stability of nonlinear systems, the nonsingular terminal sliding-mode control (NTSMC) method is developed in [23]. Furthermore, NTSMC schemes are adopted in the control of a quadrotor UAV in the presence of external disturbances (wind disturbances). In [24], a finite-time disturbance observer-based NTSMC is proposed for a quadrotor UAV with external disturbances. In [25], an NTSMC-based fault-tolerant control for a quadrotor UAV is proposed with the consideration of model uncertainties and wind disturbances. In [26], an adaptive nonsingular fast TSMC is adopted to stabilize the attitude of a quadrotor UAV, while a robust backstepping sliding-mode control is used to follow the prescribed path.

Motivated by the aforementioned analysis, this article proposes an active wind rejection control scheme for a quadrotor UAV based on the NTSMC, which explicitly considers the wind effects acting on the quadrotor dynamics. A two-stage particle filter (TSPF) is first designed for the estimation of UAV states and wind conditions from the motion of the quadrotor UAV without additional wind sensors. Then, by combining an adaptive wind compensation with the baseline NTSMC-based controller, the active wind rejection control scheme is developed to attenuate the external wind disturbances. The main contributions of this article are summarized as follows.

1) An active wind rejection control scheme is proposed for a quadrotor UAV to accommodate unknown wind disturbances and system noises. The proposed scheme can obtain the estimation of UAV states and winds from the TSPF, which can take a better response to the rapidly changing wind conditions and reduce the noise effects in the control system.
2) A TSPF is adopted to estimate the UAV states and wind conditions from quadrotor motion without additional wind sensors. The proposed TSPF can obtain the estimation from the posterior distribution of samplings, thus solving the state and parameter estimation problem for the nonlinear and non-Gaussian system of a quadrotor UAV.
3) The adaptive drag coefficients are designed to generate appropriate control signals as the compensation for the model uncertainties and wind estimation errors.

The rest of this article is organized as follows. Section II outlines the preliminaries and problem statement. The active wind rejection control scheme design and analysis are presented in Section III. Section IV provides the simulation results to demonstrate the effectiveness of the proposed scheme. Finally, Section V outlines the conclusion and future work.

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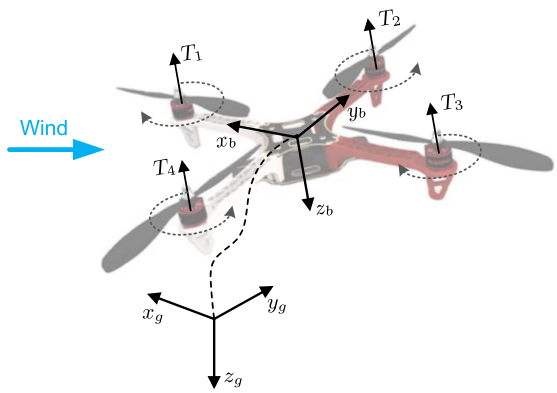

Fig. 1. Configuration of a quadrotor UAV.

II. PRELIMINARIES AND PROBLEM FORMULATION

A. Quadrotor Dynamic Model With Wind Drags

Fig. 1 shows the general configuration of a quadrotor UAV. The motion of the vehicle is controlled by the thrusts of four rotors T_i $(i = 1, \ldots, 4)$. In addition to the propulsion system, the drag force of the airframe is one of the main influential aerodynamic forces that affects the motion of the vehicle. The drag force can be approximated as the functions of the relative airspeed, i.e., the difference between the wind speed and the ground speed of the vehicle in the literature [27]. Since the quadrotor UAV operates at low translational speed, the drag forces are approximated as

$$\begin{cases} F_{wx} = C_{dxy}(\xi_x - \dot{x})\sqrt{(\xi_x - \dot{x})^2 + (\xi_y - \dot{y})^2} \\ F_{wy} = C_{dxy}(\xi_y - \dot{y})\sqrt{(\xi_x - \dot{x})^2 + (\xi_y - \dot{y})^2} \\ F_{wz} = C_{dz}(\xi_z - \dot{z}) \end{cases} \tag{1}$$

where (F_{wx}, F_{wy}, F_{wz}) are the drag forces related to winds, (C_{dxy}, C_{dz}) are the lateral and vertical drag coefficients, $(\dot{x}, \dot{y}, \dot{z})$ represent the ground speeds of the vehicle, and (ξ_x, ξ_y, ξ_z) are the wind components for each axis.

In general, the Newton–Euler equations are adopted to describe the motion of the quadrotor UAV for the easy understanding and expression. Nevertheless, the vehicle should be protected from aggressive operations to avoid the singularity issue. Without the consideration of the ground effect and other proximity-related phenomena, the quadrotor dynamic model with wind drags is given by [27]

$$\begin{cases} \ddot{x} = -\frac{(\cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi)F_z}{m} + \frac{F_{wx}}{m} + \frac{d_x}{m} \\ \ddot{y} = -\frac{(\cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi)F_z}{m} + \frac{F_{wy}}{m} + \frac{d_y}{m} \\ \ddot{z} = -\frac{(\cos\phi\cos\theta)F_z}{m} + g + \frac{F_{wz}}{m} + \frac{d_z}{m} \\ \ddot{\phi} = \frac{I_y - I_z}{I_x}\dot{\theta}\dot{\psi} + \frac{\tau_\phi}{I_x} + \frac{d_\phi}{I_x} \\ \ddot{\theta} = \frac{I_z - I_x}{I_y}\dot{\phi}\dot{\psi} + \frac{\tau_\theta}{I_y} + \frac{d_\theta}{I_y} \\ \ddot{\psi} = \frac{I_x - I_y}{I_z}\dot{\phi}\dot{\theta} + \frac{\tau_\psi}{I_z} + \frac{d_\psi}{I_z} \end{cases} \tag{2}$$

where (x, y, z) are the positions in the Earth-fixed coordinate system, (ϕ, θ, ψ) are the roll, pitch, and yaw angles, $(F_z, \tau_\phi, \tau_\theta, \tau_\psi)$ are the control force (lift) and moments, $(d_x, d_y, d_z, d_\phi, d_\theta, d_\psi)$ represent the unknown disturbances, which are assumed to be bounded, i.e., $|d| \leq D$, m is the mass of the quadrotor UAV, g is the gravity, and (I_x, I_y, I_z) are the moment of inertia for each axis.

REMARK 1 In this article, the drag force of the airframe is considered as the main external aerodynamic forces that may affect the motion of the vehicle. Assuming that the vehicle can maintain a stable flight with small tilt angles in the presence of unknown winds, the drag coefficients are reasonable to be set as constant values. It is assumed that the states of the quadrotor UAV can be measured by onboard sensors, such as the Global Positioning System and the Inertial Measurement Unit. The distributions of the measurement noises are assumed to be known. In addition, the actuators of the quadrotor UAV are assumed to be ideal and without delay.

B. Control Objective

The control objective of this article is to design an active wind rejection control scheme for the quadrotor UAV such that the vehicle can maintain the tracking performance in the presence of unknown wind disturbances and system noises.

III. ACTIVE WIND REJECTION CONTROL SCHEME DESIGN

In this section, an active wind rejection control scheme is designed for a quadrotor UAV. The framework of the proposed scheme is illustrated in Fig. 2. In the design of the proposed control scheme, a TSPF-based estimator is first developed to estimate the UAV states and unknown wind conditions from quadrotor motion with onboard sensors. Then, an active wind rejection controller is designed to attenuate the wind effects based on the estimated wind conditions.

A. TSPF for UAV State and Wind Estimation

In order to facilitate the design of the TSPF for the estimation of UAV states and wind disturbances, the artificial wind evolution is modeled by the random walk process. Therefore, the stochastic model of the quadrotor UAV is given by

$$\begin{cases} \xi_k \sim p(\xi_k|\xi_{k-1}) \\ \chi_k \sim p(\chi_k|\chi_{k-1}, \xi_k) \\ \zeta_k \sim p(\zeta_k|\chi_k) \end{cases} \tag{3}$$

where ξ_k, χ_k, and ζ_k represent the wind vector, the state vector, and the measurement vector of a quadrotor UAV at the time instant t_k, respectively, which are defined as

$$\begin{cases} \xi_k = [\xi_{x,k}, \xi_{y,k}, \xi_{z,k}]^T \\ \chi_k = [x_k, y_k, z_k, \dot{x}_k, \dot{y}_k, \dot{z}_k, \\ \qquad \phi_k, \theta_k, \psi_k, \dot{\phi}_k, \dot{\theta}_k, \dot{\psi}_k]^T \\ \zeta_k = [\zeta_{x,k}, \zeta_{y,k}, \zeta_{z,k}, \zeta_{\dot{x},k}, \zeta_{\dot{y},k}, \zeta_{\dot{z},k}, \\ \qquad \zeta_{\phi,k}, \zeta_{\theta,k}, \zeta_{\psi,k}, \zeta_{\dot{\phi},k}, \zeta_{\dot{\theta},k}, \zeta_{\dot{\psi},k}]^T. \end{cases} \tag{4}$$

The structure of the proposed TSPF is demonstrated in Fig. 3, which is composed of the first-stage state estimation and the second-stage wind estimation. The UAV state is chosen as an intermediate variable to establish a relationship

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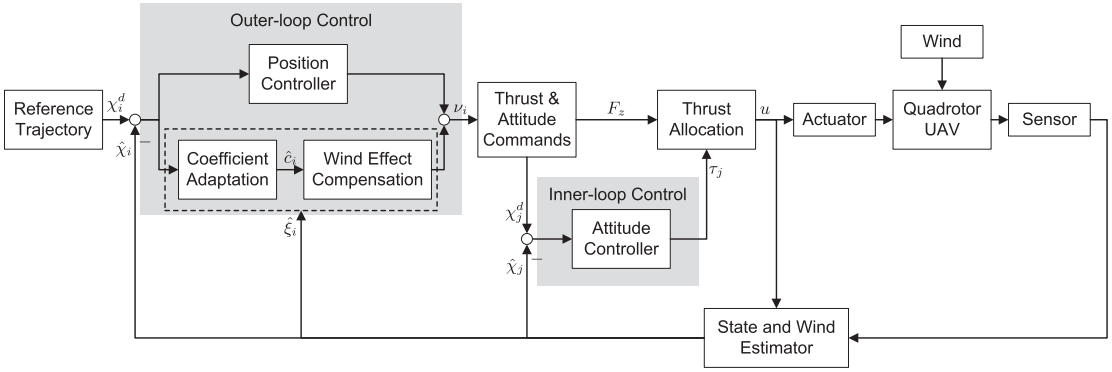

Fig. 2. Framework of the proposed active wind rejection control scheme.

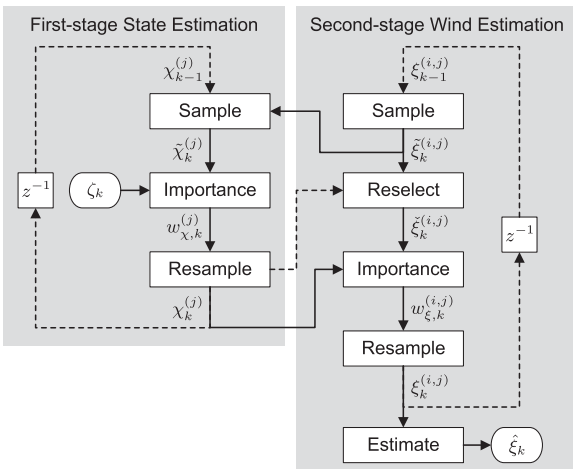

Fig. 3. Structure of the TSPF for UAV state and wind estimation.

between the winds and the measurements.

$$p(\xi_k|\zeta_{1:k}) = \int p(\xi_k|\chi_k)p(\chi_k|\zeta_{1:k})\mathrm{d}\chi_k \tag{5}$$

where $\zeta_{1:k} = \{\zeta_1, \ldots, \zeta_k\}$.

The first-stage state estimation is designed to solve the problem of estimating $p(\chi_k|\zeta_{1:k})$

$$\begin{aligned} &p(\chi_k|\zeta_{1:k}) \\ &\propto p(\zeta_k|\chi_k)\int p(\chi_k|\chi_{k-1})p(\chi_{k-1}|\zeta_{1:k-1})\mathrm{d}\chi_{k-1}. \end{aligned} \tag{6}$$

Consider that a set of state particles $\{\chi_{k-1}^{(j)}\}_{j=1}^M$ and M sets of wind particles $\{\xi_{k-1}^{(i,j)}\}_{i=1}^N$ for $j = 1, 2, \ldots, M$ are obtained at the time t_{k-1}, where M stands for the number of particles in the first-stage filter and N stands for the number of particles in each subfilter of the second-stage filter. Sampling $\tilde{\xi}_k^{(i,j)} \sim p(\xi_k|\xi_{k-1}^{(i,j)})$ and $\tilde{\chi}_k^{(j)} \sim \frac{1}{N}\sum_{i=1}^N p(\chi_k|\chi_{k-1}^{(j)}, \tilde{\xi}_k^{(i,j)})$, $p(\chi_k|\zeta_{1:k})$ can be approximated as

$$p(\chi_k|\zeta_{1:k}) \approx \sum_{j=1}^M w_{\chi,k}^{(j)}\delta(\chi_k - \tilde{\chi}_k^{(j)}) \tag{7}$$

where $\delta(\cdot)$ is the Dirac delta mass and $w_{\chi,k}^{(j)}$ is the normalized weight that is calculated as

$$w_{\chi,k}^{(j)} = \frac{p(\zeta_k|\tilde{\chi}_k^{(j)})}{\sum_{m=1}^M p(\zeta_k|\tilde{\chi}_k^{(m)})}. \tag{8}$$

After resampling with the weights, the equally weighted state particles $\{\chi_k^{(j)}\}_{j=1}^M$ can be obtained.

The second-stage wind estimation contains M subfilters, each of which is designed to solve the problem of estimating $p(\xi_k|\chi_k^{(j)})$

$$p\left(\xi_k|\chi_k^{(j)}\right) \propto p\left(\chi_k^{(j)}|\xi_k,\ \chi_{k-1}^{(j)}\right)p\left(\xi_k|\chi_{k-1}^{(j)}\right). \tag{9}$$

Reselecting the set of wind particles $\{\check{\xi}_k^{(i,j)}\}_{i=1}^N$ for each subfilter according to $\chi_k^{(j)}$, $p(\xi_k|\chi_k^{(j)})$ can be approximated as

$$p(\xi_k|\chi_k^{(j)}) \approx \sum_{i=1}^N w_{\xi,k}^{(i,j)}\delta(\xi_k - \check{\xi}_k^{(i,j)}) \tag{10}$$

where $w_{\xi,k}^{(i,j)}$ is the normalized weight that is calculated as

$$w_{\xi,k}^{(i,j)} = \frac{p(\chi_k^{(j)}|\chi_{k-1}^{(j)}, \check{\xi}_k^{(i,j)})}{\sum_{n=1}^N p(\chi_k^{(j)}|\chi_{k-1}^{(j)}, \check{\xi}_k^{(n,j)})}. \tag{11}$$

After resampling with the weights, the equally weighted state particles $\{\xi_k^{(i,j)}\}_{i=1}^N$ can be obtained.

Therefore, recalling (5), $p(\xi_k|\zeta_{1:k})$ can be approximated as

$$\begin{aligned} p(\xi_k|\zeta_{1:k}) &= \int p(\xi_k|\chi_k)p(\chi_k|\zeta_{1:k})\mathrm{d}\chi_k \\ &\approx \frac{1}{M}\sum_{j=1}^M p(\xi_k|\chi_k^{(j)}) \\ &\approx \frac{1}{MN}\sum_{i=1}^N\sum_{j=1}^M \delta(\xi_k - \xi_k^{(i,j)}). \end{aligned} \tag{12}$$

Subsequently, the wind estimation can be calculated as

$$\hat{\xi}_k \approx \frac{1}{MN}\sum_{i=1}^N\sum_{j=1}^M \xi_k^{(i,j)}. \tag{13}$$

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Algorithm 1: Proposed TSPF for the UAV State and Wind Estimation.

Input: Measurement of UAV motion ζ_k
Output: Estimate of wind vector $\hat{\xi}_k$

Initialize state particles $\chi_0^{(j)} \sim p(\chi_0)$ and wind particles $\xi_0^{(i,j)} \sim p(\xi_0)$;
for $k \leftarrow 1$ **to** *Number of time steps* **do**
  Sample $\tilde{\xi}_k^{(i,j)} \sim p(\xi_k|\xi_{k-1}^{(i,j)})$;
  Sample $\tilde{\chi}_k^{(j)} \sim \frac{1}{N}\sum_{i=1}^{N} p(\chi_k|\chi_{k-1}^{(j)}, \tilde{\xi}_k^{(i,j)})$;
  Importance $w_{\chi,k}^{(j)} \leftarrow \frac{p(\zeta_k|\tilde{\chi}_k^{(j)})}{\sum_{m=1}^{M} p(\zeta_k|\tilde{\chi}_k^{(m)})}$;
  Resample $\{\chi_k^{(j)}\}_{j=1}^{M}$ from $\{\tilde{\chi}_k^{(j)}\}_{j=1}^{M}$ with weight $\{w_{\chi,k}^{(j)}\}_{j=1}^{M}$;
  for *each subfilter* j **do**
    Reselect $\{\check{\xi}_k^{(i,j)}\}_{i=1}^{N}$ according to the resamples $\chi_k^{(j)}$;
    Importance $w_{\xi,k}^{(i,j)} \leftarrow \frac{p(\chi_k^{(j)}|\chi_{k-1}^{(j)}, \check{\xi}_k^{(i,j)})}{\sum_{n=1}^{N} p(\chi_k^{(j)}|\chi_{k-1}^{(j)}, \check{\xi}_k^{(n,j)})}$;
    Resample $\{\xi_k^{(i,j)}\}_{i=1}^{N}$ from $\{\check{\xi}_k^{(i,j)}\}_{i=1}^{N}$ with weight $\{w_{\xi,k}^{(i,j)}\}_{i=1}^{N}$;
  end
  Estimate $\hat{\xi}_k \approx \frac{1}{MN}\sum_{i=1}^{N}\sum_{j=1}^{M} \xi_k^{(i,j)}$
end

To better understand the proposed TSPF for the UAV state and wind estimation, the pseudocode of the proposed TSPF is given in Algorithm 1. Compared with the Kalman filter-based methods, the proposed TSPF-based method can avoid truncation errors due to model linearization. In addition, the TSPF-based method can better solve the sampling problem of non-Gaussian systems, which is still a great challenge for the Kalman filter-based methods.

REMARK 2 In this article, inspired by the early paper for the non-Gaussian state estimation with fading measurements [28], a TSPF is presented to estimate the UAV states and unknown winds for a quadrotor UAV according to the posterior distribution of the particles. Based on the augmented state particle filter, this TSPF-based approach separates the estimation of the UAV states and unknown winds into the first-stage filter and the second-stage filter. In theory, the estimates of $\hat{\chi}_k$ and $\hat{\xi}_k$ almost surely converge to $\mathbb{E}(\chi_k|\zeta_{1:k})$ and $\mathbb{E}(\xi_k|\zeta_{1:k})$ as $M \to \infty$. Nevertheless, there still seems to be a lack strict proof that, for unbounded functions, the approximation result converges to the true optimal estimate as the number of particles tends to infinity [29], [30].

B. Active Wind Rejection Controller

As shown in Fig. 2, the proposed hierarchical control system is mainly composed of the outer-loop control and inner-loop control. The outer-loop control is designed to generate virtual control inputs to maintain the tracking performance of a quadrotor UAV in the presence of winds, while the inner-loop control is designed to produce control moments to stabilize the attitude of the vehicle. The NTSMC method is chosen as the baseline control method in this article, because it can effectively solve the singular problem in the TSMC [23]. It should be noted that the proposed active wind rejection scheme can be also composed with other baseline control methods.

1) *Outer-Loop Control:* The quadrotor UAV is a typical underactuated system because the number of outputs is greater than the number of control inputs. To solve this underactuating problem, the virtual control inputs are defined as

$$\begin{cases} \nu_x = -(\cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi)F_z \\ \nu_y = -(\cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi)F_z \\ \nu_z = -(\cos\phi\cos\theta)F_z \end{cases} \tag{14}$$

where ν_i for $i \in \{x, y, z\}$ denotes the virtual control input.

Therefore, based on (2), the outer-loop dynamics of a quadrotor UAV can be rewritten as

$$\begin{cases} \dot{\chi}_{i,1} = \chi_{i,2} \\ \dot{\chi}_{i,2} = G_i + h_i(\nu_i + c_i\mu_i + d_i) \end{cases} \tag{15}$$

where $\chi_{x,1} = x$, $\chi_{y,1} = y$, $\chi_{z,1} = z$, $\chi_{x,2} = \dot{x}$, $\chi_{y,2} = \dot{y}$, $\chi_{z,2} = \dot{z}$, $G_x = G_y = 0$, $G_z = g$, $h_x = h_y = h_z = \frac{1}{m}$, $c_x = c_y = C_{dxy}$, $c_z = C_{dz}$, $\mu_x = (\xi_x - \dot{x})\sqrt{(\xi_x - \dot{x})^2 + (\xi_y - \dot{y})^2}$, $\mu_y = (\xi_y - \dot{y})\sqrt{(\xi_x - \dot{x})^2 + (\xi_y - \dot{y})^2}$, and $\mu_z = \xi_z - \dot{z}$.

Let χ_i^d for $i \in \{x, y, z\}$ denote the desired trajectory. Then, the trajectory tracking error can be defined as

$$\tilde{\chi}_i = \chi_{i,1} - \chi_i^d \tag{16}$$

and the derivative of the trajectory tracking error is given by

$$\dot{\tilde{\chi}}_i = \dot{\chi}_{i,1} - \dot{\chi}_i^d = \chi_{i,2} - \dot{\chi}_i^d. \tag{17}$$

By combining (15) and (17), the error dynamics can be obtained as

$$\begin{aligned} \ddot{\tilde{\chi}}_i &= \dot{\chi}_{i,2} - \ddot{\chi}_i^d \\ &= G_i + h_i(\nu_i + c_i\mu_i + d_i) - \ddot{\chi}_i^d. \end{aligned} \tag{18}$$

Introduce the following nonsingular terminal sliding-mode surface [23]:

$$s_i = \tilde{\chi}_i + \frac{1}{\beta_i}\dot{\tilde{\chi}}_i^{\frac{p_i}{q_i}} \tag{19}$$

where β_i, p_i, and q_i are positive parameters. Specifically, p_i and q_i are positive odd numbers, which satisfies $1 < \frac{p_i}{q_i} < 2$.

To allow the sliding variable to reach the designated sliding surface and then remain in close proximity to the sliding surface, the position control law can be designed in the following form:

$$\nu_i = \nu_i^{\text{eq}} + \nu_i^{\text{dis}} \tag{20}$$

where ν_i^{eq} is the equivalent control part to stabilize the ideal system without considering the disturbances, and ν_i^{dis} is the discontinuous control part to compensating for the disturbances.

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Without considering the wind drags, the equivalent control law is designed as

$$\nu_i^{\mathrm{eq}} = -\frac{1}{h_i}\left(\beta_i \frac{q_i}{p_i}\dot{\tilde{\chi}}_i^{2-\frac{p_i}{q_i}} + G_i - \ddot{\chi}_i^d\right). \tag{21}$$

Then, to cope with the disturbances, the discontinuous control law is designed as

$$\nu_i^{\mathrm{dis}} = -\eta_i \mathrm{sign}(s_i) \tag{22}$$

where $\eta_i > D_i$ is positive parameters.

In this way, the NTSMC-based outer-loop control law without wind drag compensation is designed as

$$\nu_i^o = -\frac{1}{h_i}\left(\beta_i \frac{q_i}{p_i}\dot{\tilde{\chi}}_i^{2-\frac{p_i}{q_i}} + G_i - \ddot{\chi}_i^d\right) - \eta_i \mathrm{sign}(s_i). \tag{23}$$

With the consideration of model uncertainties, the control law to compensate for the wind drags is designed as

$$\nu_i^c = -\hat{c}_i \mu_i \tag{24}$$

where $\hat{c}_i$ for $i \in \{x, y, z\}$ denote the estimation of drag coefficients. The corresponding adaptive law to estimate the drag coefficients is given by

$$\dot{\hat{c}}_i = \gamma_i s_i \frac{1}{\beta_i}\frac{p_i}{q_i}\dot{\tilde{\chi}}_i^{\frac{p_i}{q_i}-1} h_i \mu_i \tag{25}$$

Therefore, the ultimate outer-loop control law to generate the virtual control signals is given by

$$\begin{aligned}\nu_i &= \nu_i^o + \nu_i^c \\ &= -\frac{1}{h_i}\left(\beta_i \frac{q_i}{p_i}\dot{\tilde{\chi}}_i^{2-\frac{p_i}{q_i}} + G_i - \ddot{\chi}_i^d\right) - \eta_i \mathrm{sign}(s_i) - \hat{c}_i \mu_i\end{aligned} \tag{26}$$

where $i \in \{x, y, z\}$.

Considering the wind estimation errors produced by the TSPF, the error associated with the drag forces is denoted as $\tilde{\mu}_i^{-1} = \mu_i^{-1} - \hat{\mu}_i^{-1}$. Then, the model (15) can be reconstructed as

$$\begin{aligned}\dot{\chi}_{i,2} &= G_i + h_i(\nu_i + c_i\mu_i + d_i) \\ &= G_i + h_i[\nu_i + (1 - \mu_i\tilde{\mu}_i^{-1})c_i\hat{\mu}_i + d_i] \\ &= G_i + h_i(\nu_i + c_i^*\hat{\mu}_i + d_i)\end{aligned} \tag{27}$$

where c_i^* for $i \in \{x, y, z\}$ are defined as the new drag coefficients of the reconstructed model. By employing the adaptive law to estimate the newly defined drag coefficients, one can compensate for the wind estimation errors to maintain the tracking performance.

REMARK 3 The disturbance observer is usually adopted to estimate the lumped disturbances on the quadrotor UAV, and the corresponding control law is developed to attenuate the aerodynamic drag forces. However, this article develops the active wind rejection control law for a quadrotor UAV based on the wind estimates obtained by the proposed TSPF. Compared with the disturbance observer-based control scheme, the proposed control scheme can have an accurate and quick transient response to the rapidly changing winds and attenuate the influence of the system noises on the quadrotor system.

Based on the virtual control signals, the control force and desired roll and pitch angles can be computed by

$$\begin{cases} F_z = \sqrt{\nu_x^2 + \nu_y^2 + \nu_z^2} \\ \chi_\phi^d = \arcsin\left(-\frac{\nu_x \sin\chi_\psi^d - \nu_y \cos\chi_\psi^d}{\sqrt{\nu_x^2+\nu_y^2+\nu_z^2}}\right) \\ \chi_\theta^d = \arctan\left(\frac{\nu_x \cos\chi_\psi^d + \nu_y \sin\chi_\psi^d}{\nu_z}\right). \end{cases} \tag{28}$$

2) *Inner-Loop Control:* Based on (2), the inner-loop dynamics of a quadrotor UAV can be rewritten as

$$\begin{cases} \dot{\chi}_{j,1} = \chi_{j,2} \\ \dot{\chi}_{j,2} = f_j + h_j(\tau_j + d_j) \end{cases} \tag{29}$$

where $\chi_{\phi,1} = \phi$, $\chi_{\theta,1} = \theta$, $\chi_{\psi,1} = \psi$, $\chi_{\phi,2} = \dot{\phi}$, $\chi_{\theta,2} = \dot{\theta}$, $\chi_{\psi,2} = \dot{\psi}$, $f_\phi = \frac{I_y - I_z}{I_x}\dot{\theta}\dot{\psi}$, $f_\theta = \frac{I_z - I_x}{I_y}\dot{\phi}\dot{\psi}$, $f_\psi = \frac{I_x - I_y}{I_z}\dot{\phi}\dot{\theta}$, $h_\phi = \frac{1}{I_x}$, $h_\theta = \frac{1}{I_y}$, and $h_\psi = \frac{1}{I_z}$.

With the desired attitudes χ_j^d for $j \in \{\phi, \theta, \psi\}$ obtained from (28), the attitude tracking error is defined as

$$\tilde{\chi}_j = \chi_{j,1} - \chi_{j,1}^d \tag{30}$$

and the derivative of the attitude tracking error is given by

$$\dot{\tilde{\chi}}_j = \dot{\chi}_{j,1} - \dot{\chi}_{j,1}^d = \chi_{j,2} - \dot{\chi}_j^d. \tag{31}$$

By combining (29) and (31), the error dynamics can be obtained as

$$\begin{aligned}\ddot{\tilde{\chi}}_j &= \dot{\chi}_{j,2} - \ddot{\chi}_j^d \\ &= f_j + h_j(\tau_j + d_j) - \ddot{\chi}_j^d.\end{aligned} \tag{32}$$

Again, select the nonsingular terminal sliding-mode surface [23]

$$s_j = \tilde{\chi}_j + \frac{1}{\beta_j}\dot{\tilde{\chi}}_j^{\frac{p_j}{q_j}} \tag{33}$$

where β_j, p_j, and q_j are positive design parameters. p_j and q_j are odd numbers, which satisfies $1 < \frac{p_j}{q_j} < 2$.

Therefore, the attitude control law is designed as

$$\begin{aligned}\tau_j &= \tau_j^{\mathrm{eq}} + \tau_j^{\mathrm{dis}} \\ &= -\frac{1}{h_j}\left(\beta_j \frac{q_j}{p_j}\dot{\tilde{\chi}}_j^{2-\frac{p_j}{q_j}} + f_j - \ddot{\chi}_j^d\right) - \eta_j \mathrm{sign}(s_j)\end{aligned} \tag{34}$$

where $\eta_j > D_j$ are positive design parameters.

C. Stability Analysis

LEMMA 1 Consider the system state trajectories converge toward the sliding surface. The η-reachability condition is a strong condition that ensures an ideal sliding motion in finite time, even in the presence of external disturbances or uncertainties [31]. The expression of the η-reachability is given by

$$s\dot{s} \le -\eta|s| \tag{35}$$

where η represents a positive design scalar.

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THEOREM 1 Consider the outer-loop dynamic model of a quadrotor UAV (15) in the presence of wind drags and bounded disturbances. Then, the desired trajectory tracking performance can be guaranteed if the wind estimation scheme is designed as (6)–(13), the adaptive law to the drag coefficient is designed as (25), and the outer-control law is designed as (26).

PROOF Consider the following Lyapunov candidate function:

$$V_P = \frac{1}{2}\sum_{i=x}^{z} s_i^2 + \frac{1}{\gamma_i}\tilde{c}_i^2 \tag{36}$$

where $\tilde{c}_i = \hat{c}_i - c_i$ is the estimation error of drag coefficient for $i \in \{x, y, z\}$.

By differentiating (19) with respect to time, the expression can be obtained as

$$\begin{aligned}\dot{s}_i &= \dot{\tilde{\chi}}_i + \frac{1}{\beta_i}\frac{p_i}{q_i}\dot{\tilde{\chi}}_i^{\frac{p_i}{q_i}-1}\ddot{\tilde{\chi}}_i \\ &= \dot{\tilde{\chi}}_i + \frac{1}{\beta_i}\frac{p_i}{q_i}\dot{\tilde{\chi}}_i^{\frac{p_i}{q_i}-1}[G_i + h_i(v_i + c_i\mu_i + d_i) - \ddot{\chi}_i^d].\end{aligned} \tag{37}$$

Thus, the time derivative of the Lyapunov candidate function is obtained as

$$\begin{aligned}\dot{V}_P &= \sum_{i=x}^{z} s_i\dot{s}_i + \frac{1}{\gamma_i}\tilde{c}_i\dot{\hat{c}}_i \\ &= \sum_{i=x}^{z} s_i\left\{\dot{\tilde{\chi}}_i + \frac{1}{\beta_i}\frac{p_i}{q_i}\dot{\tilde{\chi}}_i^{\frac{p_i}{q_i}-1}\left[G_i + h_i(v_i + c_i\mu_i \right.\right. \\ &\quad \left.\left. + d_i) - \ddot{\chi}_i^d\right]\right\} + \frac{1}{\gamma_i}\tilde{c}_i\dot{\hat{c}}_i.\end{aligned} \tag{38}$$

Substituting the adaptive law (25) and the control law (26) into (38), the derivative of the selected Lyapunov candidate function can be calculated as

$$\begin{aligned}\dot{V}_P &= \sum_{i=x}^{z} s_i\left\{\dot{\tilde{\chi}}_i + \frac{1}{\beta_i}\frac{p_i}{q_i}\dot{\tilde{\chi}}_i^{\frac{p_i}{q_i}-1}\left[G_i \right.\right. \\ &\quad + h_i\left(-\frac{1}{h_i}\left(\beta_i\frac{q_i}{p_i}\dot{\tilde{\chi}}_i^{2-\frac{p_i}{q_i}} + G_i - \ddot{\chi}_i^d\right) - \eta_i\mathrm{sign}(s_i)\right. \\ &\quad \left.\left. - \hat{c}_i\mu_i + c_i\mu_i + d_i\right) - \ddot{\chi}_i^d\right]\Big\} + \frac{1}{\gamma_i}\tilde{c}_i\dot{\hat{c}}_i \\ &= \sum_{i=x}^{z} s_i\frac{1}{\beta_i}\frac{p_i}{q_i}\dot{\tilde{\chi}}_i^{\frac{p_i}{q_i}-1}[-h_i\hat{c}_i\mu_i + h_ic_i\mu_i + h_id_i \\ &\quad - h_i\eta_i\mathrm{sign}(s_i)] + \frac{1}{\gamma_i}\tilde{c}_i\dot{\hat{c}}_i \\ &= \sum_{i=x}^{z}\tilde{c}_i\left(\frac{1}{\gamma_i}\dot{\hat{c}}_i - s_i\frac{1}{\beta_i}\frac{p_i}{q_i}\dot{\tilde{\chi}}_i^{\frac{p_i}{q_i}-1}h_i\mu_i\right) \\ &\quad + \frac{1}{\beta_i}\frac{p_i}{q_i}\dot{\tilde{\chi}}_i^{\frac{p_i}{q_i}-1}h_i(-\eta_is_i\mathrm{sign}(s_i) + d_is_i) \\ &\le \sum_{i=x}^{z} -\eta_i'|s_i|\end{aligned} \tag{39}$$

TABLE I
Parameters of the Quadrotor Dynamic Model

Parameter	Value	Unit
m	1.5	kg
g	9.8	N/kg
I_x	0.03	$\mathrm{kg}\cdot\mathrm{m}^2$
I_y	0.03	$\mathrm{kg}\cdot\mathrm{m}^2$
I_z	0.05	$\mathrm{kg}\cdot\mathrm{m}^2$
C_{dxy}	0.3	-
C_{dz}	0.5	-

where $\eta_i' = \frac{1}{\beta_i}\frac{p_i}{q_i}\dot{\tilde{\chi}}_i^{\frac{p_i}{q_i}-1}h_i(\eta_i - D_i) \ge 0$.

Therefore, the system satisfies the standard η-reachability condition to guarantee the trajectory tracking performance of the designed outer-loop controller in the presence of wind drags and bounded disturbances. ■

REMARK 4 It is important to notice that due to the application of the discontinuous controller v_i^{dis}, the usual regularity conditions for the unicity of closed-loop solutions do not hold, which is an in-depth topic. Nevertheless, in this article, it is sufficient to note that the only discontinuity in the closed-loop system is caused by sign(·), which means that the limited solutions exist and they exist for all time, implying that the closed-loop system is forward complete [32], [33]. To deal with the chattering and discontinuous problems caused by the sign(·) function, the hyperbolic tangent function tanh(·) can be used as an alternative. The hyperbolic tangent function is defined as

$$\tanh\left(\frac{s}{\epsilon}\right) = \frac{e^{\frac{s}{\epsilon}} - e^{-\frac{s}{\epsilon}}}{e^{\frac{s}{\epsilon}} + e^{-\frac{s}{\epsilon}}} \tag{40}$$

where $\epsilon > 0$ is the parameter.

IV. SIMULATION RESULTS

In this section, simulations are conducted to demonstrate the effectiveness of the proposed active wind rejection control scheme for a quadrotor UAV against unknown winds. The parameters of the dynamic model of the quadrotor UAV are given in Table I.

The initial states of the quadrotor UAV are set as $x(0) = 0$ m, $y(0) = 0$ m, $z(0) = 0$ m, $\phi(0) = 0$ rad, $\theta(0) = 0$ rad, and $\psi(0) = 0$ rad. The controller design parameters are set as $\beta_x = \beta_y = \beta_z = 1$, $p_x = p_y = p_z = 13$, $q_x = q_y = q_z = 11$, $\eta_x = \eta_y = \eta_z = 0.5$, $\beta_\phi = \beta_\theta = \beta_\psi = 5$, $p_\phi = p_\theta = p_\psi = 13$, $q_\phi = q_\theta = q_\psi = 11$, and $\eta_\phi = \eta_\theta = \eta_\psi = 1$. The parameters of the adaptive law (25) are set as $(\gamma_x, \gamma_y, \gamma_z) = (1, 1, 1)$. The particle numbers of the first-stage and second-stage particle filters are set as $M = 200$ and $N = 200$, respectively.

To get closer to the actual situation, the system noises are considered. The process noise is assumed to be independent zero-mean Gaussian noise, the variance of which is set as

$$Q_k = \begin{bmatrix} 10^{-6}I_{6\times6} & \\ & 10^{-12}I_{6\times6}\end{bmatrix} \tag{41}$$

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where Q_k denotes the covariance matrix of the process noise. The measurement noise is modeled by a mixture of two Gaussian distributions

$$p(\zeta_k|\chi_k) = \alpha\mathcal{N}(\mu_1, \Sigma_1) + (1-\alpha)\mathcal{N}(\mu_2, \Sigma_2) \tag{42}$$

where α is the glint presence probability. In the simulation, the mean vectors of the two Gaussian distributions are set as $\mu_1 = 0_{12\times1}$ and $\mu_2 = 10^{-1}I_{12\times1}$, and the covariance matrices of the two Gaussian distributions are set as

$$\begin{cases} \Sigma_1 = \begin{bmatrix} 10^{-2}I_{6\times6} & \\ & 10^{-4}I_{6\times6} \end{bmatrix} \\ \Sigma_2 = \begin{bmatrix} 10^{-3}I_{6\times6} & \\ & 10^{-5}I_{6\times6} \end{bmatrix}. \end{cases} \tag{43}$$

It should be noticed that the Kalman filter-based approaches may not work perfectly or even diverge due to the non-Gaussian measurement noises.

The random walk model is adopted to propagate the wind particles as

$$\xi_k = \xi_{k-1} + w_{\xi,k} \tag{44}$$

where $w_{\xi,k}$ denotes the zero-mean Gaussian process noise vector, the variances of which are set as $Q_{\xi,k} = \mathrm{diag}([0.01,\ 0.01,\ 0.01])$.

To demonstrate the advantages of the active wind rejection control scheme for a quadrotor UAV against unknown winds, a nonlinear disturbance observer (NDO)-based control scheme is adopted to compare with the proposed control scheme. The NDO can estimate the unknown slow time-varying lumped external disturbance and generate control input compensation for a quadrotor UAV by considering the difference between the measurements and the states of the nominal plant model [11]. In principle, the designs of NDO-based control scheme and the proposed control scheme both depend on the dynamic model of a quadrotor UAV. They have the comparable performances against unknown slow time-varying winds when there is no measurement noise. However, the influences of measurement noises on the performance of the control schemes should be considered in practical engineering applications. According to (15), the NDO is expressed as

$$\begin{cases} \hat{d}_i = \varpi_i + L_i \cdot \chi_{i,2} \\ \dot{\varpi}_i = -L_i \cdot \hat{d}_i - L_i \cdot \left(G_i + \frac{1}{m}v_i\right) \end{cases} \tag{45}$$

where ϖ_i for $i \in \{x,\ y,\ z\}$ denotes the internal state of the nonlinear observer and L_i is the NDO gain.

A. Hovering Performance With No Wind

In the first scenario, the quadrotor UAV is controlled to reach and hover at the position $[1,\ 1,\ 1]^T$ m from the initial position in the absence of external wind disturbances. The performance of the vehicle is affected by the system noises. Fig. 4 shows the state estimation results of the proposed TSPF method. It can be seen that the proposed TSPF method can provide the accurate estimation of the UAV states, although the measurement noise is non-Gaussian.

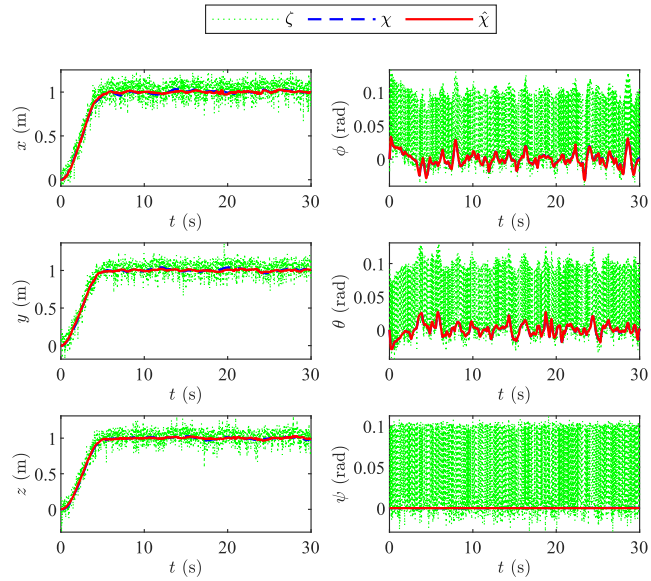

Fig. 4. State estimation performance in Scenario 1.

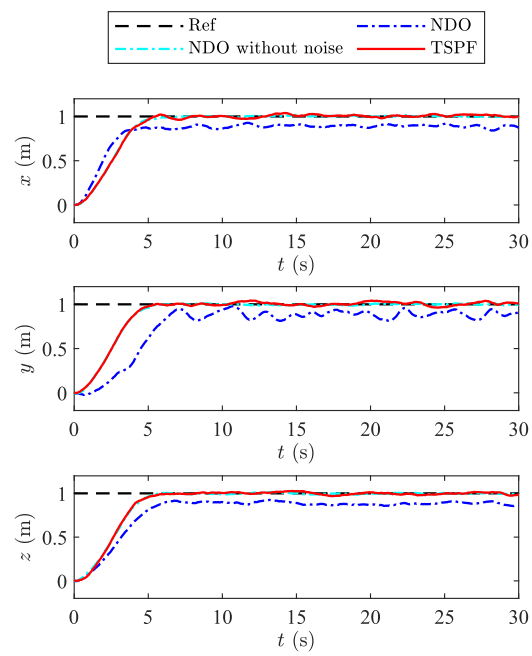

Fig. 5. Comparison of the hovering performance in Scenario 1.

Fig. 5 demonstrates the comparison of the hovering performance in Scenario 1. It can be observed that the NDO-based control method has the poor performance in the presence of nonzero-mean measurement noises. It is because the disturbance observation is considerably influenced by measurement noises. As a result, the control input compensation generated by NDO includes the component of measurement noises. It can also be seen that there is a considerable improvement in the performance of the NDO-based control scheme without considering the measurement noises. The simulation results also illustrate that the proposed TSPF-based control scheme can reduce the noise effects and maintain the hovering performance of the quadrotor UAV in the presence of system noises. It is

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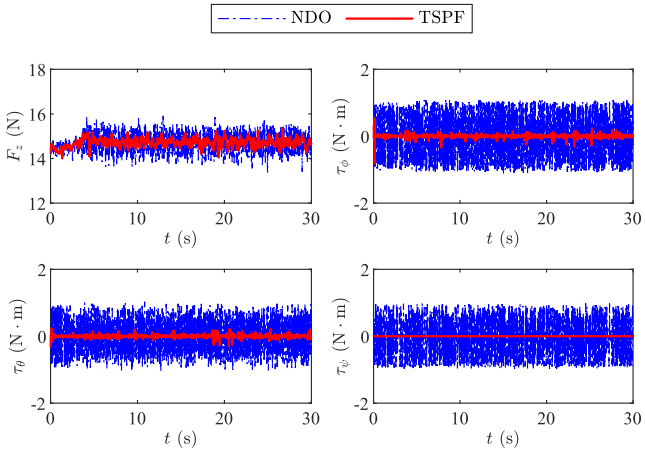

Fig. 6. Comparison of the control signals in Scenario 1.

because TSPF can search for a set of particles to approximate the distribution of UAV states from the posterior distribution in a stochastic problem formulation, while NDO and other DO-based approaches are basically deterministic approaches. That is also the reason and new contributions toward such an application for us to develop the TSPF-based approach in this article.

Fig. 6 shows the comparison of the control signals in Scenario 1. It can be seen that due to the existence of system noises, the signals of the control force and moments, which are generated by the NTSMC-based controller, oscillate significantly. Compared with the NDO-based control method, the proposed TSPF-based control scheme can effectively attenuate the oscillations by reducing the system noise effects. Nevertheless, in the practical engineering applications, the control signals are usually smoothed by the low-pass filter before they are sent to the actuators.

B. Hovering Performance Against Gust Winds

In the second scenario, the quadrotor UAV is controlled to hover against gust winds. The gust winds are rendered according to the discrete 1-cosine wind gust model [34], which are expressed as

$$\xi_x = \begin{cases} 0, & t \le 10 \\ \frac{4}{2}(1-\cos\frac{2\pi(t-10)}{5}), & 10 < t \le 15 \\ 0, & t > 15 \end{cases} \tag{46}$$

$$\xi_y = \begin{cases} 0, & t \le 10 \\ \frac{3}{2}(1-\cos\frac{2\pi(t-10)}{5}), & 10 < t \le 15 \\ 0, & t > 15 \end{cases} \tag{47}$$

$$\xi_z = \begin{cases} 0, & t \le 10 \\ \frac{2}{2}(1-\cos\frac{2\pi(t-10)}{5}), & 10 < t \le 15 \\ 0, & t > 15. \end{cases} \tag{48}$$

Figs. 7 and 8 show the state and wind estimation results in Scenario 2. It can be seen that the proposed TSPF-based method can effectively reduce the system noises and accurately capture the rapid changes of wind gusts.

Fig. 9 shows the comparison of the hovering performance in Scenario 2. When the wind gusts occur at the time

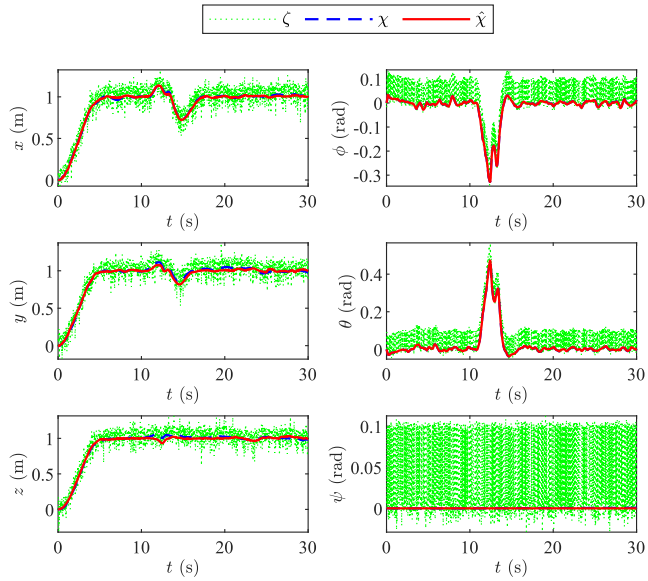

Fig. 7. State estimation performance in Scenario 2.

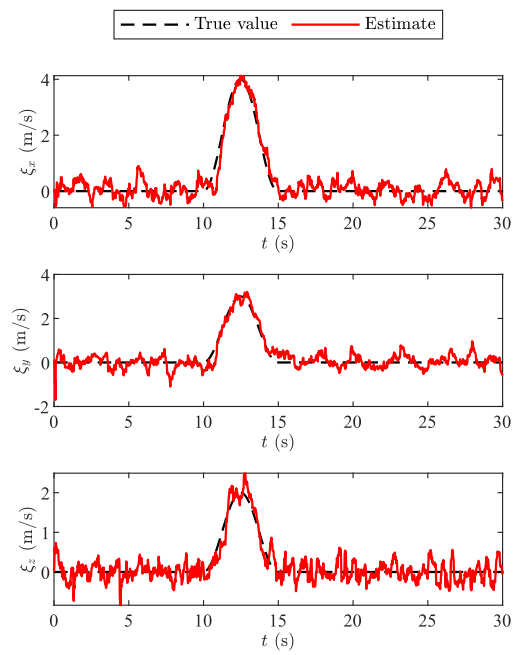

Fig. 8. Wind estimation performance in Scenario 2.

instant $t = 10$ s, there are significant position drifts of the quadrotor UAV by using the NDO-based control method. This is because the disturbance observer (45) cannot take a quick response to the rapidly changing wind disturbances. In contrast, the proposed TSPF-based control scheme can actively attenuate the effects of wind gusts based on the wind estimation provided by the TSPF. From Fig. 10, it can be seen that the control signals are given by the proposed control scheme with significant oscillations to attenuate the wind effects when the wind gusts occur, although the oscillations are unattainable for the actuators.

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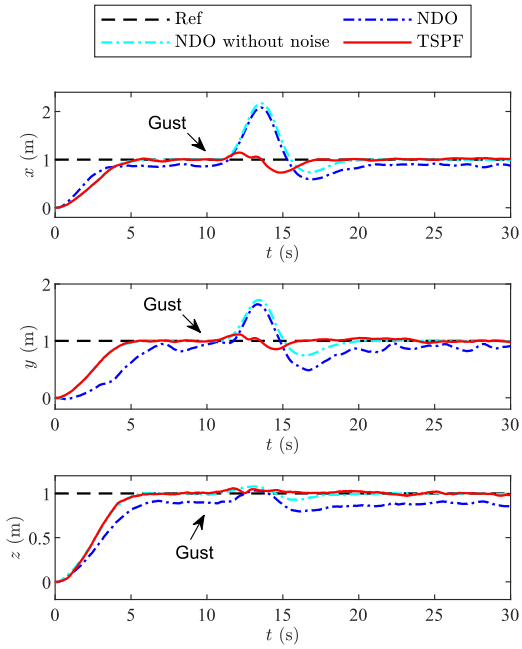

Fig. 9. Comparison of the hovering performance in Scenario 2.

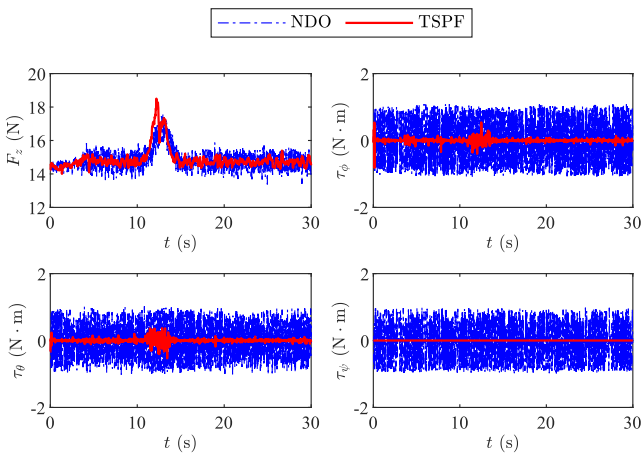

Fig. 10. Comparison of the control signals in Scenario 2.

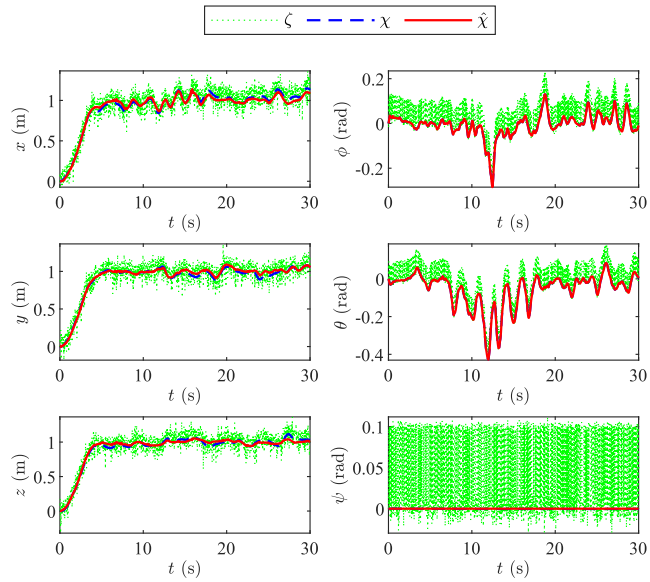

Fig. 11. State estimation performance in Scenario 3.

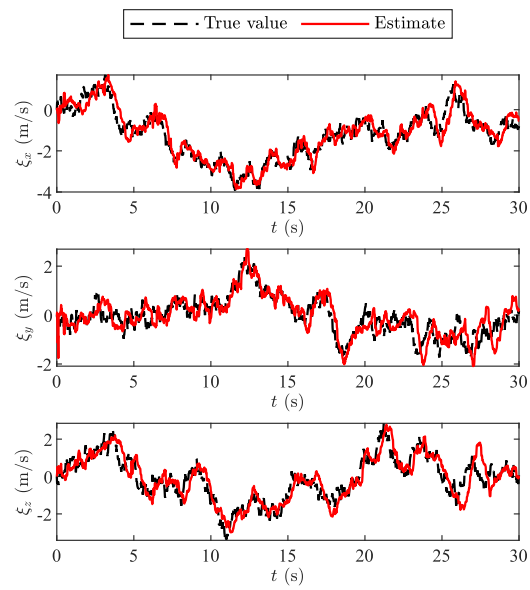

Fig. 12. Wind estimation performance in Scenario 3.

C. Hovering Performance Against Dryden Turbulence

In the third scenario, the quadrotor UAV is controlled to hover against the Dryden turbulence [34], [35]. Figs. 11 and 12 show the estimation results of the UAV states and the Dryden turbulence produced by the proposed TSPF. It can be seen that the hovering performance of the quadrotor UAV is continuously affected by the wind turbulence. The proposed TSPF method can accurately capture the continuous changes of the wind turbulence.

Fig. 13 illustrates the comparison of the hovering performance in Scenario 3. It can be seen that the proposed TSPF-based control scheme can outperform the NDO-based control methods in the presence of the Dryden turbulence. This is because the proposed scheme can provide the estimation with higher transient accuracy for the continuously changing winds. In addition, the proposed scheme can reduce the influence of the system noises to maintain the hovering performance of the vehicle. From Fig. 14, it can be observed that the wind turbulence results in the continuous oscillations in the generation of the control signals.

D. Sinusoidal Trajectory Tracking Performance Against Continuous Winds

In the fourth scenario, the quadrotor UAV is controlled to track a sinusoidal trajectory in the presence of continuous winds. The desired trajectory is given by

$$\begin{cases} x^d = 5\cos(\frac{\pi}{15}t) - 5 \\ y^d = 5\sin(\frac{\pi}{15}t) \\ z^d = -\sin(\frac{2\pi}{15}t). \end{cases} \tag{49}$$

Fig. 15 shows the comparison of the trajectory tracking performance in Scenario 4, and the trajectory tracking errors are given in Table II. The mean absolute errors and the

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TABLE II
Comparison of Trajectory Tracking Errors Against Continuous Winds

	TSPF	NDO	NDO without noise
x mean absolute error (m)	0.0501	0.1616	0.0894
y mean absolute error (m)	0.0955	0.3000	0.1649
z mean absolute error (m)	0.0387	0.1138	0.0496
x root mean squared error (m)	0.0703	0.2237	0.1225
y root mean squared error (m)	0.1901	0.4104	0.2612
z root mean squared error (m)	0.0468	0.1306	0.0664

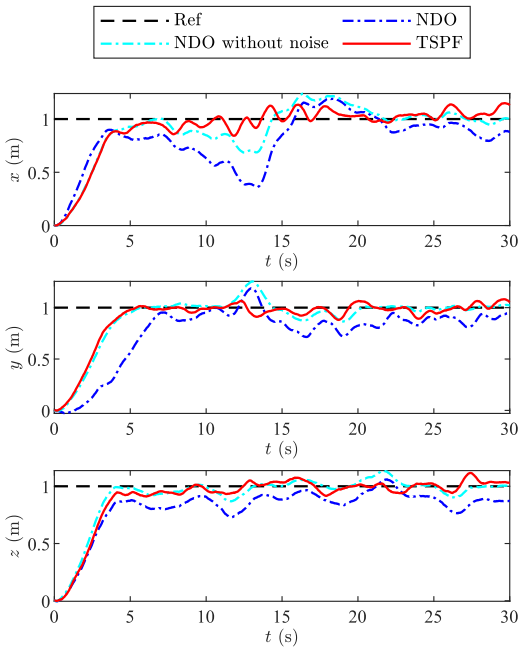

Fig. 13. Comparison of the hovering performance in Scenario 3.

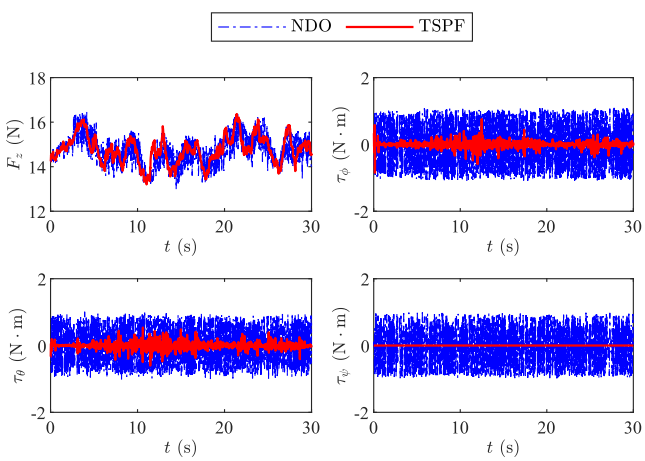

Fig. 14. Comparison of the control signals in Scenario 3.

root mean squared errors of the TSPF-based scheme are smaller than those of the NDO-based control methods. This indicates that the trajectory tracking performance of the proposed TSPF-based control scheme is also better than the NDO-based control methods.

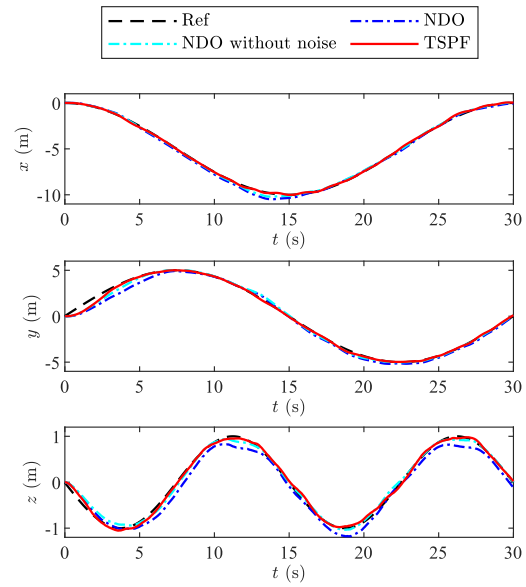

Fig. 15. Comparison of the trajectory tracking performance in Scenario 4.

V. CONCLUSION AND FUTURE WORK

In this article, an active wind rejection control scheme is proposed for a quadrotor UAV against unknown winds. The designed TSPF can quickly capture the wind variations and produce accurate wind estimation from quadrotor motion without additional wind sensors. Based on the explicit wind estimation and the baseline NTSMC, the proposed control scheme can actively generate appropriate control signals to attenuate the wind disturbances. Moreover, considering the wind estimation errors, the adaptive drag coefficients can be adopted to maintain the tracking performance of the quadrotor UAV. Simulation results have demonstrated the effectiveness of the proposed control scheme against unknown winds. Nevertheless, issues of actuator and sensor faults and model uncertainties [36] have not yet been addressed in the current work. One of the future works will be to investigate these factors, which may have a significant impact on the performance of the active wind rejection control scheme. In addition, the effectiveness of the proposed control scheme for a quadrotor UAV against unknown winds should be validated by real flight tests in the future.

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